How the validation split changes a MOF screening claim

Six-step workflow for comparing random and chemically grouped holdouts

1 Data foundation

INPUTS + GROUP LABELS

- ARC–MOF uptake targets
- Descriptors: model inputs
- Group labels: define holdouts

Four adsorption targets

2 Validation design

WHAT IS HELD OUT?

- Five random 20% test splits
- Five grouped folds
- Five group-label families

Six validation regimes

3 Model evaluation

MATCHED COMPARISON

- Matched models and metrics
- Compare two descriptor sets
- Track coverage differences

Ridge RF HGB MLP

4 Error patterns

WHAT FAILS TO TRANSFER?

- Held-out R^2 and MAE
- Errors by fold and group
- Gas and pressure dependence

Fold- and family-level evidence

5 Candidate ranking

DO THE SAME MOFs SURVIVE?

- Reference top-5% recovery
- Predicted-list Jaccard
- Common out-of-sample set

Recovery ≠ agreement

6 Evidence and limits

WHAT CAN WE CLAIM?

- Inspect difficult MOFs
- Trace cases to source data
- Define evidence boundaries

Retrospective validation evidence

Interpretation boundary A holdout tests transfer in that setting; it does not establish mechanisms or prospective success.